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Fundamentals of statistical learning

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1

Probability

1.1 Why probability?

Observed data are the results of processes whose behavior we may not fully know. To reason about them, we describe the process with a probabilistic model. Probability lets us work forward from a model to possible outcomes; statistical inference uses observed outcomes to learn about the model.

Example 1: A coin toss

Let $\theta = P(\text{heads})$ be the probability that a coin lands heads. If θ is known, we can ask how likely it is to observe three heads in five tosses. If instead we observe three heads in five tosses while θ is unknown, we can use those data to learn about θ .

1.2 Sample spaces and events

Before assigning probabilities, we must specify what the experiment can produce. We start with a finite set of possible results.

Definition 1.1: Outcomes and events

A random experiment is a process whose result is not known in advance. Each possible result is an *outcome* ω . The set of all possible outcomes is the *sample space* Ω . An *event* is a subset $A \subseteq \Omega$; it occurs when the observed outcome belongs to A .

Example 2: Rolling a die

For one roll of a six-sided die, the sample space is $\Omega = \{1, 2, 3, 4, 5, 6\}$. The event “the result is even” is $A = \{2, 4, 6\}$. The whole sample space Ω is the certain event, while $\emptyset$ is the impossible event.

1.2.1 Operations on events

Events are subsets of the same sample space Ω , so set operations let us describe new events in terms of A and B :

- **Union:** $A \cup B := \{\omega \in \Omega : \omega \in A \text{ or } \omega \in B\}$. This event occurs when at least one of A and B occurs; “or” includes the case in which both occur. In particular, $A \cup \emptyset = A$ and $A \cup \Omega = \Omega$.
- **Intersection:** $A \cap B := \{\omega \in \Omega : \omega \in A \text{ and } \omega \in B\}$. This event occurs when both A and B occur. In particular, $A \cap \Omega = A$ and $A \cap \emptyset = \emptyset$.
- **Complement:** $A^c := \Omega \setminus A = \{\omega \in \Omega : \omega \notin A\}$. This event occurs when A does not occur. In particular, $A \cup A^c = \Omega$ and $A \cap A^c = \emptyset$.

For the die in Example 2, let $B = \{4, 5, 6\}$ be the event “the result is at least four”. Then $A \cup B = \{2, 4, 5, 6\}$, $A \cap B = \{4, 6\}$, and $A^c = \{1, 3, 5\}$.

Two events are *disjoint* when $A \cap B = \emptyset$, so they cannot occur together. If $A \subseteq B$, every outcome in A is also in B : whenever A occurs, B occurs.

Proposition 1.1: De Morgan’s laws

For any events A and B ,

$$(A \cup B)^c = A^c \cap B^c, \quad (A \cap B)^c = A^c \cup B^c.$$

More generally, for events $A_1, \dots, A_n$,

$$\left(\bigcup_{i=1}^n A_i \right)^c = \bigcap_{i=1}^n A_i^c, \quad \left(\bigcap_{i=1}^n A_i \right)^c = \bigcup_{i=1}^n A_i^c.$$

The first identity says that “none occurs” means that every event fails. The second says that “not all occur” means that at least one event fails. We will use the second identity in the proof of Bonferroni’s inequality.

1.2.2 Partitions and event decomposition

A useful way to study an event is to split it into disjoint pieces. For any events A and B , exactly one of B and B^c occurs, so

$$A = A \cap \Omega = A \cap (B \cup B^c) = (A \cap B) \cup (A \cap B^c).$$

The two events on the right are disjoint. For example, a car can fail an emissions test either while it pollutes or while it does not.

More generally, a *partition* $B_1, \dots, B_n$ of Ω consists of pairwise disjoint events whose union is Ω . Every event A can then be written as

$$A = \bigcup_{i=1}^n (A \cap B_i), \quad (A \cap B_i) \cap (A \cap B_j) = \emptyset \quad (i \neq j).$$

This decomposition will let us add the probabilities of different ways in which the same event can occur.

1.3 Probability

There are several ways to interpret a probability. Under a *symmetry* assumption, equally likely outcomes give $P(A) = |A|/|\Omega|$. A *frequency* interpretation concerns the long-run proportion of repetitions in which A occurs. A *subjective* interpretation expresses a degree of belief about A . The axioms below state the mathematical rules that a probability assignment must satisfy, regardless of its interpretation.

Example 3: Two dice

If the 36 ordered results of rolling two dice are equally likely, they form the sample space $\Omega = \{1, \dots, 6\}^2$. The sum is eight for $(2, 6), (3, 5), (4, 4), (5, 3), (6, 2)$, so the probability of that event is $5/36$. Counting outcomes this way is valid because the ordered pairs are equally likely.

Definition 1.2: Probability on a finite sample space

A probability measure P assigns a number $P(A)$ to every event $A \subseteq \Omega$ such that:

1. $0 \leq P(A) \leq 1$ for every event A ;
2. $P(\Omega) = 1$;
3. if $A \cap B = \emptyset$, then $P(A \cup B) = P(A) + P(B)$.

The last property is called *finite additivity*.

Because A and A^c are disjoint and their union is Ω , these axioms give $P(A^c) = 1 - P(A)$. In particular, $P(\emptyset) = 0$.

They also imply *monotonicity*: if $A \subseteq B$, then $B = A \cup (B \cap A^c)$ is a disjoint union, so $P(B) = P(A) + P(B \cap A^c) \geq P(A)$.

Example 4: A fair die

If the six outcomes of a die roll are equally likely, each has probability $1/6$. The event $A = \{2, 4, 6\}$ therefore has probability $P(A) = 3/6 = 1/2$.

For an infinite sample space, finite additivity is replaced by *countable additivity*: for pairwise disjoint events $A_1, A_2, \dots$,

$$P\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} P(A_i).$$

In that setting, a probability measure is defined on a suitable family of events rather than necessarily on every subset of Ω . On a finite sample space, the finite formulation above is enough.

1.3.1 Inclusion–exclusion

Finite additivity directly handles disjoint events. If A and B overlap, adding $P(A)$ and $P(B)$ counts their intersection twice.

Proposition 1.2: Inclusion–exclusion for two events

For any events A and B ,

$$P(A \cup B) = P(A) + P(B) - P(A \cap B).$$

Proof. Split $A \cup B$ into the disjoint events A and $B \cap A^c$:

$$P(A \cup B) = P(A) + P(B \cap A^c).$$

The event B is itself the disjoint union of $A \cap B$ and $A^c \cap B$. Thus $P(B \cap A^c) = P(B) - P(A \cap B)$, which gives the formula. $\square$

Example 5: Overlapping die events

Let $A = \{2, 4, 6\}$ and $B = \{4, 5, 6\}$. Then $P(A) = P(B) = 1/2$ and $P(A \cap B) = 1/3$. Therefore

$$P(A \cup B) = \frac{1}{2} + \frac{1}{2} - \frac{1}{3} = \frac{2}{3},$$

in agreement with $A \cup B = \{2, 4, 5, 6\}$.

Applying the two-event formula twice gives the three-event version:

$$\begin{aligned} P(A \cup B \cup C) &= P(A) + P(B) + P(C) \\ &\quad - P(A \cap B) - P(A \cap C) - P(B \cap C) \\ &\quad + P(A \cap B \cap C). \end{aligned}$$

The final term restores outcomes in all three events: they were added three times and subtracted three times.

For n events, the same pattern continues:

$$P\left(\bigcup_{i=1}^n A_i\right) = \sum_{\emptyset \neq I \subseteq \{1, \dots, n\}} (-1)^{|I|+1} P\left(\bigcap_{i \in I} A_i\right).$$

To see why, consider an outcome contained in exactly k of the events. It is counted $\binom{k}{j}$ times among the intersections of size j , so its total coefficient on the right is $\sum_{j=1}^k (-1)^{j+1} \binom{k}{j} = 1$. An outcome in none of the events contributes zero to both sides.

1.3.2 Probability bounds

Sometimes the probabilities of intersections are unknown. We can still bound the probability of a union.

Proposition 1.3: Union bound (Boole's inequality)

For events $A_1, \dots, A_n$,

$$P\left(\bigcup_{i=1}^n A_i\right) \leq \sum_{i=1}^n P(A_i).$$

Proof. We prove the bound by induction on the number of events.

Base case ($n = 1$). The union contains only A_1 , so the inequality is an equality: $P(A_1) = \sum_{i=1}^1 P(A_i)$.

Inductive step. Assume the bound holds for n events and set $U_n = \bigcup_{i=1}^n A_i$. The union of $n + 1$ events is $U_n \cup A_{n+1}$. Applying inclusion–exclusion gives

$$\begin{aligned} P\left(\bigcup_{i=1}^{n+1} A_i\right) &= P(U_n) + P(A_{n+1}) - P(U_n \cap A_{n+1}) \\ &\leq P(U_n) + P(A_{n+1}) \\ &\leq \sum_{i=1}^n P(A_i) + P(A_{n+1}) = \sum_{i=1}^{n+1} P(A_i). \end{aligned}$$

The first inequality uses $P(U_n \cap A_{n+1}) \geq 0$; the second uses the induction hypothesis. Thus the bound holds for every $n \geq 1$. $\square$

For a countable sequence, the same inequality follows by applying the finite bound to the first n events and taking the limit.

Proposition 1.4: Bonferroni's inequality

For events $A_1, \dots, A_n$,

$$P\left(\bigcap_{i=1}^n A_i\right) \geq \sum_{i=1}^n P(A_i) - (n - 1).$$

Proof. The second n -event form of De Morgan's law turns the complement of an intersection into a union of complements. Then the union bound gives

$$\begin{aligned} P\left(\bigcap_{i=1}^n A_i\right) &= 1 - P\left(\bigcup_{i=1}^n A_i^c\right) \\ &\geq 1 - \sum_{i=1}^n P(A_i^c) \\ &= 1 - \sum_{i=1}^n (1 - P(A_i)) = \sum_{i=1}^n P(A_i) - (n - 1). \end{aligned}$$

$\square$

1.3.3 Application: a randomized identity test

Suppose F and G are polynomials of degree at most d . We want to test whether they are identical without comparing all their coefficients. Choose a nonempty finite set S of possible evaluation points.

Algorithm 1.1: Randomized polynomial identity test

Given F , G , and S , report whether the polynomials differ or may be identical.

```

function IDENTITYTEST( $F, G, S$ )
  Draw  $R$  uniformly at random from  $S$ 
  if  $F(R) \neq G(R)$  then
    return DIFFERENT
  else
    return POSSIBLY EQUAL
  end if
end function

```

If $F = G$, the test never reports a difference. If $F \neq G$, it fails to detect the difference only when R belongs to the event

$$E = \{r \in S : F(r) = G(r)\}.$$

Indeed, E contains the roots in S of the nonzero polynomial $H = F - G$, which has at most d roots. Because R is uniform on S ,

$$P(R \in E) = \frac{|E|}{|S|} \leq \frac{d}{|S|}.$$

For $S = \{1, \dots, 100d\}$ with $d \geq 1$, the probability of missing a difference is at most $1/100$. The possible error is *one-sided*: “different” is certain, while “possibly equal” is not a proof of identity.

1.4 Independent events

1.5 Conditional probability

1.6 Bayes’ theorem

1.6.1 Odds

For an event A with $P(A) < 1$, its *odds in favor* are

$$O(A) := \frac{P(A)}{P(A^c)} = \frac{P(A)}{1 - P(A)}.$$

Conversely, $P(A) = O(A)/(1 + O(A))$. For a fair die, the odds in favor of rolling a six are $(1/6)/(5/6) = 1/5$, often stated as one to five. This is different from saying that the probability is $1/5$; the probability is $1/6$.

2

Crash course exercises

2.1 Day 01

Exercise 1: Computer virus

A virus can enter a computer through e-mail or the internet. The probability of entry through e-mail is 0.30, through the internet 0.40, and through both routes 0.15. What is the probability that the virus does not enter at all?

Solution.

Let

$E = \{\text{the virus enters through e-mail}\}, \quad I = \{\text{the virus enters through the internet}\}.$

Step 1: Define the target event. Let $N = E^c \cap I^c$ be the event that the virus does not enter. We seek $P(N)$.

Step 2: Apply De Morgan's law.

$$P(N) = P((E \cup I)^c).$$

Step 3: Use the complement rule.

$$P((E \cup I)^c) = 1 - P(E \cup I).$$

Step 4: Calculate the union by inclusion–exclusion.

$$P(E \cup I) = P(E) + P(I) - P(E \cap I) = 0.30 + 0.40 - 0.15 = 0.55.$$

Step 5: Take the complement.

$$P(N) = 1 - 0.55 = 0.45 = 45\%.$$

Exercise 2: Two children

A family has two children, and at least one is a boy. What is the probability that both are boys?

Solution.

Assume each child is independently a boy or a girl with probability $1/2$. Let

$$B_1 = \{\text{the first child is a boy}\}, \quad B_2 = \{\text{the second child is a boy}\}.$$

Step 1: Define the events in the question. Let

$$\begin{aligned} A &= B_1 \cap B_2 = \{\text{both children are boys}\}, \\ C &= B_1 \cup B_2 = \{\text{at least one child is a boy}\}. \end{aligned}$$

We seek $P(A \mid C)$.

Step 2: List the ordered outcomes. With B for boy and G for girl, the equally likely possibilities are

$$\Omega = \{BB, BG, GB, GG\}.$$

Step 3: Use the information given. The event C excludes GG , leaving

$$C = \{BB, BG, GB\}.$$

Step 4: Count the favorable outcomes. We have $A = \{BB\}$, so one of the three outcomes in C is favorable:

$$P(A \mid C) = \frac{1}{3}.$$

Check using conditional probability.

Since $A \subseteq C$, we have $A \cap C = A$. Independence and the complement rule give

$$P(A) = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}, \quad P(C) = 1 - P(\{GG\}) = 1 - \frac{1}{4} = \frac{3}{4}.$$

Therefore

$$P(A \mid C) = \frac{P(A \cap C)}{P(C)} = \frac{1/4}{3/4} = \frac{1}{3}.$$

If the information had instead been *the first child is a boy*, only BB and BG would remain, giving $1/2$. The way the information is supplied matters.

2.2 Day 02

Exercise 3: Rolling two dice

Two fair six-sided dice are rolled. Describe the sample space and find the probability that their sum is seven.

Solution.

Step 1: Write the sample space. The result is an ordered pair (i, j) , one value for each die:

$$\Omega = \{(i, j) : i, j \in \{1, \dots, 6\}\}, \quad |\Omega| = 6 \cdot 6 = 36.$$

In particular, $(1, 6)$ and $(6, 1)$ are different outcomes.

Step 2: Define the target event. Let

$$S = \{(i, j) \in \Omega : i + j = 7\} = \{(1, 6), (2, 5), (3, 4), (4, 3), (5, 2), (6, 1)\}.$$

We seek $P(S)$, and $|S| = 6$.

Step 3: Apply the equally likely outcomes rule.

$$P(S) = \frac{|S|}{|\Omega|} = \frac{6}{36} = \frac{1}{6}.$$

Exercise 4: Five-card poker hands

Assume that all five-card hands from a standard 52-card deck are equally likely. Find the probabilities of a flush (five cards of one suit), four of a kind, and exactly one pair.

Solution.

Because order does not matter, all three probabilities have the same denominator:

$$N = \binom{52}{5} = 2\,598\,960.$$

Flush.

Step 1: Choose the suit: 4 possibilities.

Step 2: Choose five cards of that suit: $\binom{13}{5}$ possibilities.

Step 3: Count and divide:

$$P(\text{flush}) = \frac{4 \binom{13}{5}}{N} = \frac{5\,148}{2\,598\,960} \approx 0.001981.$$

Under the exercise's definition, this includes straight flushes.

Four of a kind.

Step 1: Choose the rank: 13 possibilities. All four cards of that rank are then fixed.

Step 2: Choose the fifth card: any of the remaining 48.

Step 3: Count and divide:

$$P(\text{four of a kind}) = \frac{13 \cdot 48}{N} = \frac{624}{2\,598\,960} \approx 0.0002401.$$

Exactly one pair.

Step 1: Choose the pair's rank: 13 possibilities.

Step 2: Choose its two suits: $\binom{4}{2}$ possibilities.

Step 3: Choose the other three ranks: $\binom{12}{3}$ ways. They must be distinct, avoiding another pair or a triple.

Step 4: Choose their suits: 4^3 possibilities.

Step 5: Count and divide:

$$P(\text{exactly one pair}) = \frac{13 \binom{4}{2} \binom{12}{3} 4^3}{N} = \frac{1\,098\,240}{2\,598\,960} \approx 0.42257.$$

2.3 Day 04 (24 sept 26)

Exercise 5: Rome and Florence

The probability that a tourist has visited Rome is 26%, the probability that they have visited Florence is 18%, and the probability that they have visited at least one of the two cities is 38%.

Question A. What is the probability that the tourist has visited both Rome and Florence?

Question B. What is the probability that the tourist has visited Rome but not Florence?

Notation.

Let

$$R = \{\text{the tourist has visited Rome}\}, \quad F = \{\text{the tourist has visited Florence}\}.$$

The given probabilities are

$$P(R) = 0.26, \quad P(F) = 0.18, \quad P(R \cup F) = 0.38.$$

Solution to A.

Step 1: Define the target event. Let

$$A = R \cap F = \{\text{the tourist has visited both cities}\}.$$

We seek $P(A)$.

Step 2: Apply inclusion–exclusion and substitute.

$$\begin{aligned} P(A) &= P(R) + P(F) - P(R \cup F) \\ &= 0.26 + 0.18 - 0.38 \\ &= \boxed{0.06 = 6\%}. \end{aligned}$$

Solution to B.

Step 1: Define the target event. Let

$$B = R \cap F^c = \{\text{the tourist has visited Rome but not Florence}\}.$$

We seek $P(B)$.

Step 2: Subtract the overlap from Rome.

$$\begin{aligned} P(B) &= P(R) - P(A) \\ &= 0.26 - 0.06 \\ &= \boxed{0.20 = 20\%}. \end{aligned}$$

Exercise 6: An uncommon disease

You are diagnosed with an uncommon disease. You know that there is only a 1% chance of getting it. Let

$$D = \{\text{you have the disease}\}, \quad T = \{\text{the test is positive}\}.$$

It is known that the test is imperfect, since

$$P(T \mid D) = 0.98, \quad P(T^c \mid D^c) = 0.95.$$

Question A. Given that you test positive, what is the probability that you have the disease?

Question B. You take the test again, independently of the first test conditional on whether you have the disease, and it is positive again. Given both positive results, what is the probability that you have the disease?

Solution to A.

We seek $P(D \mid T)$. The prior probabilities are $P(D) = 0.01$ and $P(D^c) = 0.99$.

Step 1: Find the false-positive rate. By the complement rule,

$$P(T \mid D^c) = 1 - P(T^c \mid D^c) = 1 - 0.95 = 0.05.$$

Step 2: Find the probability of a positive result. By the law of total probability,

$$\begin{aligned} P(T) &= P(T \mid D)P(D) + P(T \mid D^c)P(D^c) \\ &= 0.98 \cdot 0.01 + 0.05 \cdot 0.99 \\ &= 0.0593. \end{aligned}$$

Step 3: Apply Bayes' theorem.

$$P(D \mid T) = \frac{P(T \mid D)P(D)}{P(T)} = \frac{0.0098}{0.0593} \approx 16.53\%.$$

The positive test raises the probability of disease from 1% to about 16.53% under the stated assumptions.

Solution to B.

Let

$$\begin{aligned}T_1 &= \{\text{the first test is positive}\}, \\T_2 &= \{\text{the second test is positive}\}.\end{aligned}$$

We seek $P(D \mid T_1 \cap T_2)$.

Step 1: Find the likelihood of two positive results. Conditional independence gives

$$P(T_1 \cap T_2 \mid D) = 0.98^2 = 0.9604, \quad P(T_1 \cap T_2 \mid D^c) = 0.05^2 = 0.0025.$$

Step 2: Find the probability of two positive results. By the law of total probability,

$$\begin{aligned}P(T_1 \cap T_2) &= P(T_1 \cap T_2 \mid D)P(D) + P(T_1 \cap T_2 \mid D^c)P(D^c) \\&= 0.9604 \cdot 0.01 + 0.0025 \cdot 0.99 \\&= 0.012079.\end{aligned}$$

Step 3: Apply Bayes' theorem.

$$P(D \mid T_1 \cap T_2) = \frac{0.9604 \cdot 0.01}{0.012079} \approx 79.51\%.$$

The second positive result raises the probability from about 16.53% to about 79.51% under the conditional independence assumption.

3

Random variables

Start of lecture 25 September

3.1 From outcomes to data

Imagine picking one student at random from a class. Before the draw, we do not know who we will get; afterwards, we can write down that student's height, weight, and age. Probability describes the draw, while the numbers we write down are the data.

The possible outcomes form Ω , and $\omega \in \Omega$ is the outcome that actually occurs. We write $(\Omega, \mathcal{F}, P)$ for the probability space: $\mathcal{F}$ contains the events to which P assigns probabilities. For a finite Ω , as in Chapter 1, every subset can be an event.

Definition (informal): Random object

Given a probability space, a *random object* is a map

$$Z : \Omega \longrightarrow \mathcal{T},$$

where $\mathcal{T}$ is a suitable space of observations for our data-collection process.

Definition 3.1: Random object

A *random object* is a measurable* map from outcomes to possible observations:

$$Z : \Omega \longrightarrow \mathcal{T}, \quad \omega \longmapsto Z(\omega).$$

The set $\mathcal{T}$ says what kind of thing we record. It could contain numbers, vectors, images, or text messages. Once ω is known, $Z(\omega)$ is the observation we get.

*A rigorous definition of measurability is beyond the scope of this course.

Think of Z as the rule that turns the outcome into a piece of data. “Measurable” is the technical condition that lets us assign a probability to questions about that data, such as “is the selected student taller than 180 cm?”

3.2 Random variables and random vectors

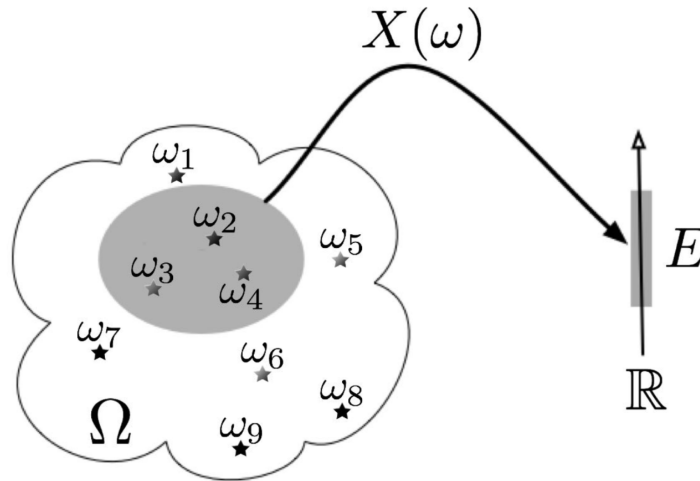
Often the observation is a number or a list of numbers. These are two particularly useful kinds of random objects.

Definition 3.2: Random variable

A *random variable* is a measurable map

$$X : \Omega \longrightarrow \mathbb{R}.$$

It assigns one real number $X(\omega)$ to each outcome ω .



A map from Ω to $\mathbb{R}$. Illustration from Brutti’s slides.

Definition 3.3: Random vector

A *p-dimensional random vector* is a measurable map

$$\underline{X} : \Omega \longrightarrow \mathbb{R}^p, \quad \underline{X}(\omega) := (X_1(\omega), \dots, X_p(\omega)).$$

Each coordinate X_j is a real-valued random variable. Thus one outcome gives us p numbers collected into a vector. These measurements can be related; they do not have to be independent.

Note: Notation

In these notes, we denote a random vector by $\underline{X}$ (X with a bar underneath):

$$\underline{X} = (X_1, \dots, X_p).$$

Example 6: One student, one number or three

Pick a student at random and let ω identify the student selected. If H records height in centimetres, then $H(\omega)$ is one number. For a student who is 178 cm tall, $H(\omega) = 178$.

Now record height H , weight W in kilograms, and age A in years for that *same* student:

$$\underline{X}(\omega) = \begin{pmatrix} H(\omega) \\ W(\omega) \\ A(\omega) \end{pmatrix} = \begin{pmatrix} 178 \\ 72 \\ 21 \end{pmatrix} \in \mathbb{R}^3.$$

The draw gives us one student, not three. We simply take three measurements of that student. Before the draw, H and $\underline{X}$ are uncertain; for the selected student, their values are fixed.

In supervised learning, each sampled item gives us an input $\underline{X}$ (the features) and a target Y (the label or quantity of interest). Before we see the item, both are uncertain; for an observed item, their values are fixed.

Example 7: House prices

Choose a house at random. Use its floor area in square metres, number of rooms, and age in years as features. For a particular house with 85 m², 3 rooms, and an age of 10 years, the data look like this:

$$\underline{X}(\omega) = (85, 3, 10), \quad Y(\omega) = 240,000 \text{ euros.}$$

The three features make a random vector; the sale price is one random variable.

Example 8: Image labels

Take a randomly chosen 64×64 grayscale photo. Each pixel is one number from 0 (black) to 255 (white). Putting the pixels in a fixed order gives $\underline{X}(\omega) \in \mathbb{R}^{4096}$: one vector with 4096 entries for one photo. If we set $Y(\omega) = 1$ for a cat and $Y(\omega) = 0$ for a dog, then a cat photo gives one feature vector and the single label 1. The pixels can be related even though they are separate coordinates of $\underline{X}$.

3.3 Preimages and induced probabilities

To ask a probability question about the values of X , we first find the outcomes that produce those values.

Definition 3.4: Preimage

For a random variable $X : \Omega \rightarrow \mathbb{R}$ and a set $E \subseteq \mathbb{R}$, the *preimage* (or *inverse image*) of E is

$$X^{-1}(E) := \{\omega \in \Omega : X(\omega) \in E\} \subseteq \Omega.$$

To find this event, we collect the outcomes whose outputs fall in E . Whenever $X^{-1}(E)$ is an event, we define

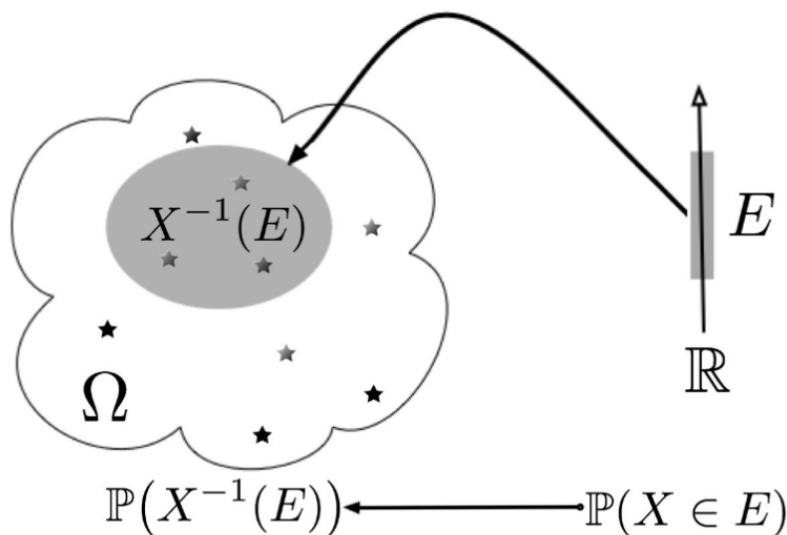
$$\begin{aligned} P(X \in E) &:= P(X^{-1}(E)) \\ &= P(\{\omega \in \Omega : X(\omega) \in E\}). \end{aligned}$$

In particular, for a single value x ,

$$P(X = x) := P(X^{-1}(\{x\})).$$

Here X is the random variable, while x is a particular real number.

The set E contains values; $X^{-1}(E)$ contains the outcomes that give those values. In our finite examples, every subset of Ω is an event.



A preimage and its probability. Illustration from Brutti's slides.

Example 9: Which students have heights in an interval?

Choose one student uniformly from $\Omega = \{\text{Anna, Luca, Sara, Marco}\}$. Let X record height in centimetres:

ω	Anna	Luca	Sara	Marco
$X(\omega)$	165	178	178	190

For $E = [170, 180]$, Luca and Sara are the students whose heights fall inside the interval. Thus

$$X^{-1}([170, 180]) = \{\text{Luca, Sara}\}, \quad P(170 \leq X \leq 180) = \frac{2}{4} = \frac{1}{2}.$$

The same two students also give the exact value 178: $X^{-1}(\{178\}) = \{\text{Luca, Sara}\}$, so $P(X = 178) = 1/2$. One value can correspond to several outcomes.

Example 10: Counting heads in two coin tosses

Toss a fair coin twice, with independent tosses. The four equally likely outcomes are $\Omega = \{HH, HT, TH, TT\}$, where H means heads and T means tails. Let X count the heads:

$$X(HH) = 2, \quad X(HT) = X(TH) = 1, \quad X(TT) = 0.$$

To get exactly one head, we need either HT or TH . Therefore

$$X^{-1}(\{1\}) = \{HT, TH\}, \quad P(X = 1) = P(\{HT, TH\}) = \frac{2}{4} = \frac{1}{2}.$$

Example 11: Counting tails in five coin tosses

Toss a fair coin five times, with independent tosses. Write H for heads and T for tails. The sample space is $\Omega = \{H, T\}^5$, containing $2^5 = 32$ sequences. Let X count the tails in a sequence, so its possible values are 0, 1, 2, 3, 4, 5.

Each particular sequence has probability $(1/2)^5 = 1/32$. For example, $HHHHH$ and $TTHHH$ are equally likely, although they give different values of X .

No tails. Only $HHHHH$ gives zero tails:

$$\begin{aligned} P(X = 0) &= P(X^{-1}(\{0\})) \\ &= P(\{HHHHH\}) \\ &= \binom{5}{0} \left(\frac{1}{2}\right)^5 = \frac{1}{32}. \end{aligned}$$

One tail. The tail can occupy any of the five positions:

$$\begin{aligned} P(X = 1) &= P(X^{-1}(\{1\})) \\ &= P(\{TTHHH, HTTHH, HHTTH, HHHHT, HHHHT\}) \\ &= \binom{5}{1} \left(\frac{1}{2}\right)^5 = \frac{5}{32}. \end{aligned}$$

Two tails. Choose which two of the five positions contain tails. The corresponding event contains $\binom{5}{2} = 10$ sequences:

$$\begin{aligned} P(X = 2) &= P(X^{-1}(\{2\})) \\ &= P(\{TTTHH, THTTH, \dots, HHHHT\}) \\ &= \binom{5}{2} \left(\frac{1}{2}\right)^5 = \frac{10}{32}. \end{aligned}$$

In general, exactly x tails can occur in $\binom{5}{x}$ sequences. Summing their probabilities gives

$$P(X = x) = P(X^{-1}(\{x\})) = \binom{5}{x} \left(\frac{1}{2}\right)^5, \quad x \in \{0, 1, 2, 3, 4, 5\},$$

The two factors are the number of sequences with x tails and the probability of each sequence.

Example 12: Darts: distance from the centre

Throw a dart at a circular target of radius $r > 0$. Assume that the dart always hits the disc and that its hitting point is uniform over the surface: regions with the same area have the same probability of being hit.

Place the centre at the origin. An outcome is the hitting point $\omega = (u, v)$, so the sample space is

$$\Omega = \{(u, v) \in \mathbb{R}^2 : u^2 + v^2 \leq r^2\}.$$

Let X measure the distance from that point to the centre:

$$X : \Omega \rightarrow \mathbb{R}, \quad X(u, v) = \sqrt{u^2 + v^2}.$$

For a target of radius 10 cm, some possible outcomes and their distances are:

Hitting point ω	Distance $X(\omega)$ (cm)
(0, 0)	0
(3, 4)	$\sqrt{3^2 + 4^2} = 5$
(-3, 4)	$\sqrt{(-3)^2 + 4^2} = 5$
(0, 10)	10

In general, the possible values of X fill the interval $[0, r]$: zero at the centre, r on the boundary, and any distance in between.

For the 10 cm target, asking for a distance of at most 5 cm selects the outcomes

$$X^{-1}([0, 5]) = \{(u, v) \in \Omega : u^2 + v^2 \leq 25\}.$$

This is the smaller disc of radius 5 cm around the centre.

Now choose a distance threshold x . For $0 \leq x \leq r$, the event $X \leq x$ means that the dart hits the inner disc of radius x . Since the hitting point is uniform over the target, its probability is the ratio of the two areas:

$$P(X \leq x) = \frac{\text{area of the inner disc}}{\text{area of the target}} = \frac{\pi x^2}{\pi r^2} = \frac{x^2}{r^2}.$$

A negative threshold is impossible for a distance, while a threshold larger than r includes the entire target. Thus

$$P(X \leq x) = \begin{cases} 0, & x < 0, \\ x^2/r^2, & 0 \leq x \leq r, \\ 1, & x > r. \end{cases}$$

For the target of radius 10 cm,

$$P(X \leq 5) = \frac{\pi \cdot 5^2}{\pi \cdot 10^2} = \frac{25}{100} = \frac{1}{4}.$$

3.4 Discrete random variables

Definition 3.5: Discrete random variable and PMF

A random variable X is *discrete* if there is a finite or countably infinite set $\mathcal{X} = \{x_1, x_2, \dots\}$ such that $P(X \in \mathcal{X}) = 1$.

Its *probability mass function* (PMF) is

$$p_X(x) := P(X = x), \quad x \in \mathbb{R}.$$

1. $p_X(x) \geq 0$ for every $x \in \mathbb{R}$.
2. The *support*, in the discrete sense used here, is $\text{supp}(X) := \{x \in \mathbb{R} : p_X(x) > 0\}$. Thus the support lists the values that have positive probability; values outside it have probability zero.
3. $\sum_{x \in \mathcal{X}} p_X(x) = 1$.

By countable additivity, for any set E ,

$$P(X \in E) = \sum_{x \in E \cap \text{supp}(X)} p_X(x).$$

Example 13: The PMF of a fair die

Roll a fair six-sided die and let X be the number shown. The random variable X is discrete because the finite set $\mathcal{X} = \{1, 2, 3, 4, 5, 6\}$ satisfies $P(X \in \mathcal{X}) = 1$.

Its *probability mass function* (PMF) is:

$$p_X(x) = \begin{cases} 1/6, & x \in \{1, 2, 3, 4, 5, 6\}, \\ 0, & \text{otherwise.} \end{cases}$$

It satisfies $p_X(x) \geq 0$ for every x , and $\sum_{x \in \mathcal{X}} p_X(x) = 6 \cdot (1/6) = 1$.

Its *support* is $\text{supp}(X) = \{1, 2, 3, 4, 5, 6\}$, since each of these values has positive probability. For example, $p_X(3) = 1/6$, whereas $p_X(7) = 0$ and 7 is outside the support.

Let's see an example that shows how to use the PMF to calculate the probability of an event. To find the probability of rolling an even number, we add the probabilities of rolling 2, 4, or 6:

$$P(X \in \{2, 4, 6\}) = p_X(2) + p_X(4) + p_X(6) = \frac{1}{6} + \frac{1}{6} + \frac{1}{6} = \frac{1}{2}.$$

*A finite set has finitely many elements. A set A is countably infinite if there exists a bijection $f : \mathbb{N} \rightarrow A$.

3.5 Continuous random variables

Definition 3.6: Continuous random variable and PDF

A random variable X is *continuous* (in the sense used here) if there exists a function f_X , called its *probability density function* (PDF), such that, for every $a \leq b$,

$$P(a \leq X \leq b) = \int_a^b f_X(x) dx.$$

1. $f_X(x) \geq 0$ for every $x \in \mathbb{R}$.
2. Following the convention in the slides, the *support* is $\text{supp}(X) := \{x \in \mathbb{R} : f_X(x) > 0\}$.
3. $\int_{\mathbb{R}} f_X(x) dx = 1$.

Example 14: Waiting for a bus

Suppose a bus arrives every 10 minutes and you reach the stop at a uniformly random time between two arrivals. Let X be your waiting time in minutes. Intervals of the same length have the same probability, so the PDF is constant on $[0, 10]$. Its height is $1/10$, since the total area must be 1.

Its PDF is:

$$f_X(x) = \begin{cases} 1/10, & 0 \leq x \leq 10, \\ 0, & \text{otherwise.} \end{cases}$$

It satisfies $f_X(x) \geq 0$ for every x , and $\int_{\mathbb{R}} f_X(x) dx = \int_0^{10} (1/10) dx = 1$.

Its *support*, using the convention above, is $[0, 10]$.

Let's see an example that shows how to use the PDF to calculate the probability of an event. The probability of waiting between 2 and 5 minutes is:

$$P(2 \leq X \leq 5) = \int_2^5 \frac{1}{10} dx = \frac{5-2}{10} = 0.30.$$

3.6 Cumulative distribution function (CDF)

Definition 3.7: Cumulative distribution function

Given a random variable X , its *cumulative distribution function* (CDF) is the function $F_X : \mathbb{R} \rightarrow [0, 1]$ defined by

$$F_X(x) := P(X \leq x) = P(X \in (-\infty, x]), \quad x \in \mathbb{R}.$$

It gives the probability accumulated up to and including the threshold x . It is defined for both [discrete](#) and [continuous](#) random variables.

Proposition: The CDF determines the distribution

The CDF contains all the information about the *distribution* of a random variable. If $F_X(t) = F_Y(t)$ for every $t \in \mathbb{R}$, then X and Y have the same distribution: for any set of values E defining an event,

$$P(X \in E) = P(Y \in E).$$

This does not mean that X and Y must produce the same outcome in an experiment.

Proof. We prove only a special case of this statement: we show that it holds for intervals $E = (a, b]$. The full result also holds for any set E defining an event, but we do not prove that general case here.

Suppose $F_X(t) = F_Y(t)$ for every $t \in \mathbb{R}$, and let $a < b$. Since

$$\{X \leq b\} = \{X \leq a\} \cup \{a < X \leq b\},$$

and the two events on the right are disjoint, additivity gives

$$P(a < X \leq b) = P(X \leq b) - P(X \leq a) = F_X(b) - F_X(a).$$

Applying the same reasoning to Y and using the equality of the two CDFs, we obtain

$$\begin{aligned} P(X \in E) &= F_X(b) - F_X(a) \\ &= F_Y(b) - F_Y(a) \\ &= P(Y \in E). \end{aligned}$$

□

Note: PMF and CDF

The [PMF](#) answers: “What is the probability that X is exactly x ?”

$$p_X(x) = P(X = x).$$

The [CDF](#) answers: “What is the probability that X is at most x ?”

$$F_X(x) = P(X \leq x).$$

3.6.1 Properties of the CDF

Proposition 3.1: Properties of the CDF (I)

A function $F(\cdot)$ mapping the real line to $[0, 1]$ is a CDF for some probability measure $P(\cdot)$ if and only if:

1. $F(\cdot)$ is *non-decreasing*: $x_1 < x_2$ implies that $F(x_1) \leq F(x_2)$.
2. $F(\cdot)$ is *normalized*:

$$\lim_{x \rightarrow -\infty} F(x) = 0, \quad \lim_{x \rightarrow +\infty} F(x) = 1.$$

3. $F(\cdot)$ is *right-continuous*: for all x we have

$$F(x) = F(x^+), \quad F(x^+) := \lim_{t \downarrow x} F(t).$$

Proposition 3.2: Properties of the CDF (II/a)

Let $F_X(\cdot)$ be the CDF of a random variable X . Then:

1. For $x_1 < x_2$,

$$P(x_1 < X \leq x_2) = P(X \in (x_1, x_2]) = F_X(x_2) - F_X(x_1).$$

In the discrete case, be careful with the strict and non-strict inequalities.

- 2.

$$P(X > x) = P(X \in (x, +\infty)) = 1 - F_X(x).$$

3. If X is continuous with PDF $f_X(\cdot)$, then

$$F_X(x) = \int_{-\infty}^x f_X(t) dt.$$

A PDF can be recovered from the CDF as

$$f_X(x) = F'_X(x)$$

at points where F_X is differentiable.

Proposition 3.3: Properties of the CDF (II/b)

Let $F_X(\cdot)$ be the CDF of a random variable X . Then:

1.
$$P(x_1 < X \leq x_2) = P(X \in (x_1, x_2]) = F_X(x_2) - F_X(x_1), \quad x_1 < x_2.$$
2.
$$P(X > x) = P(X \in (x, +\infty)) = 1 - F_X(x).$$
3. If X is discrete with PMF $p_X(\cdot)$ and support $\mathcal{X} = \{x_1, x_2, \dots\}$, then

$$F_X(x) = \sum_{x_i \leq x} p_X(x_i) = P(X < x) + p_X(x).$$

Hence

$$p_X(x) = F_X(x) - F_X(x^-), \quad F_X(x^-) := \lim_{t \uparrow x} F_X(t).$$

Let $F_X(\cdot)$ be the CDF of a continuous random variable with PDF $f_X(\cdot)$. The following informal observations motivate later nonparametric statistical developments:

1. Roughly speaking, $F_X(\cdot)$ is “simpler” than $f_X(\cdot)$.
2. It must be non-decreasing, so its shape is more constrained than that of a PDF.
3. Integration accumulates and smooths the PDF, making the CDF more regular.
4. This suggests that estimating the CDF is easier than estimating the PDF: fewer observations may be needed to reach the same statistical accuracy.

Example 15: Five coin tosses: PMF and CDF

Return to five independent flips of a balanced coin, and let X count the tails. Its possible values are 0, 1, 2, 3, 4, 5.

Its *probability mass function* (PMF) is:

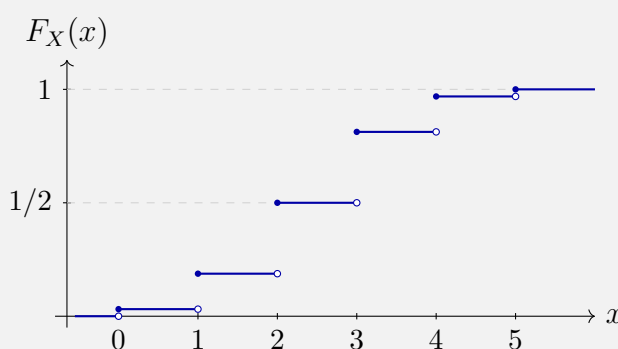
$$p_X(x) = P(X = x) = \binom{5}{x} \left(\frac{1}{2}\right)^5, \quad x \in \{0, 1, 2, 3, 4, 5\},$$

and $p_X(x) = 0$ otherwise. The probabilities are:

x	0	1	2	3	4	5
$p_X(x)$	1/32	5/32	10/32	10/32	5/32	1/32

Its *cumulative distribution function* (CDF) is obtained by adding the probabilities of all values at most x :

$$F_X(x) = \sum_{k \leq x} p_X(k) = \begin{cases} 0, & x < 0, \\ 1/32, & 0 \leq x < 1, \\ 6/32, & 1 \leq x < 2, \\ 16/32, & 2 \leq x < 3, \\ 26/32, & 3 \leq x < 4, \\ 31/32, & 4 \leq x < 5, \\ 1, & x \geq 5. \end{cases}$$



Filled points give the value of the CDF at each jump; open points are excluded.

The graph displays the three properties in [Properties of the CDF \(I\)](#):

1. *Non-decreasing*: each jump adds a positive probability.
2. *Normalized*: $F_X(x) = 0$ for $x < 0$ and $F_X(x) = 1$ for $x \geq 5$.
3. *Right-continuous*: for example, consider $x = 2$. We have $F_X(2) = 16/32 = 1/2$, and the same value holds for $2 \leq x < 3$. Thus the right limit equals $F_X(2)$.

Example 16: Darts: from areas to the CDF and PDF

Return to the [earlier darts example](#), with a disc of radius 10 cm and a hitting point uniform over its area. Let X be the distance from the centre in centimetres. Earlier, we obtained $P(X \leq x)$ using areas. Now we recognise this function as the CDF, differentiate it to find a PDF, and use that PDF to calculate probabilities of intervals.

1. Recognise the CDF. The function calculated earlier is the CDF:

$$F_X(x) = P(X \leq x) = \begin{cases} 0, & x < 0, \\ x^2/100, & 0 \leq x \leq 10, \\ 1, & x > 10. \end{cases}$$

2. Derive the PDF. The PDF is not given. We find it by differentiating the CDF for $0 < x < 10$:

$$f_X(x) = F'_X(x) = \frac{d}{dx} \left(\frac{x^2}{100} \right) = \frac{2x}{100} = \frac{x}{50}.$$

Its PDF is:

$$f_X(x) = \begin{cases} x/50, & 0 \leq x \leq 10, \\ 0, & \text{otherwise.} \end{cases}$$

This PDF reproduces the probabilities of the model. It is non-negative and satisfies

$$\int_{\mathbb{R}} f_X(x) dx = \int_0^{10} \frac{x}{50} dx = \left[\frac{x^2}{100} \right]_0^{10} = 1.$$

Its support, with the convention used above and this choice of PDF, is $(0, 10]$.

3. Use the PDF to calculate an interval probability. Now that we have derived the PDF, we can, for example, calculate the probability that the dart lands between 5 and 8 cm from the centre:

$$P(5 \leq X \leq 8) = \int_5^8 \frac{x}{50} dx = \left[\frac{x^2}{100} \right]_5^8 = \frac{64 - 25}{100} = 0.39.$$

The CDF gives the same result: $F_X(8) - F_X(5) = 0.64 - 0.25 = 0.39$.

For the two intervals used in the slides, the same method gives

$$P(0 < X \leq 5) = \int_0^5 \frac{x}{50} dx = \frac{1}{4}, \quad P(5 < X \leq 10) = \int_5^{10} \frac{x}{50} dx = \frac{3}{4}.$$

Fine lezione 25 settembre